

Heterogeneous Interface Substitution in Finite Resolution Systems: Episode Constitution and Exact Preservation of Reachable Terminal Profiles

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Abstract

Finite systems may receive resolution-relevant information through heterogeneous interface realizations. Local admissibility of individual interface outputs does not ensure that the available witnesses jointly constitute an instance of the declared resolution problem, and replacing one realization by another may alter global behavior even when the replacement appears locally compatible. We study both issues for finite guarded resolvers. A resolution contract specifies required roles, guarded transitions, and two designated terminal classes. Interface realizations produce typed post-transduction witnesses; an episode is constituted only when at least one role-complete assignment of locally admissible witnesses also satisfies declared joint structural constraints, with both checks independent of resolver outcomes. For each role, a resolver-indexed contextual equivalence identifies values indistinguishable by every relevant guard and terminal predicate in every context of the remaining roles. Three sufficient pre-resolver conditions—local contextual equivalence, preservation of constitutive assignments, and completeness of realized local-class vectors—then imply equality of realized class-vector sets and global resolution-signature sets, canonical isomorphism of the semantic quotient resolvers, and exact preservation of the reachable terminal profile. Eight finite adversarial scenarios isolate the hypotheses, while a

canonical scaling family reduces 4^k constitutive assignments to 2^k semantic signatures exactly.

The conditions are sufficient, not necessary, by design.

Keywords: heterogeneous interfaces; finite-state systems; component substitution; contextual equivalence; compositional verification; formal systems

1 Introduction

Finite decision and resolution mechanisms may depend on information supplied through heterogeneous interfaces. A required role may be instantiated by a sensor, a database, a software transformation, a learned model, or a declared composition of several sources. This heterogeneity creates two distinct formal questions. First, do the available interface outputs jointly constitute the input structure required by the declared resolution problem? Second, when one interface realization is replaced by another, which conditions are sufficient to preserve the resolver’s exact reachable terminal behavior?

These questions must be kept separate. A witness can be locally admissible yet structurally incompatible with the witnesses available for other roles. Conversely, several locally admissible witnesses for the same role may disagree while each participates in a legitimate joint assignment. Resolving that disagreement before the resolver is instantiated can delete a branch that the declared semantics is required to retain. Substitution raises a different difficulty: equality of raw values is generally stronger than necessary, whereas agreement in a single observed context is generally too weak. The relevant criterion is whether the replacement preserves every distinction that the declared resolver can observe.

We therefore separate the interface boundary from the resolver semantics. Throughout this paper, a *resolver* is the declared finite guarded mechanism instantiated only after episode constitution. The primitive interface object is a *post-transduction witness*: a typed value together with the metadata required to assess its declared provenance, reference, units, and integrity. For each required role, a local verifier evaluates an individual witness without consulting values supplied for other roles, resolver guards, terminal states, preferred outcomes, or the final reachable profile. Joint structural compatibility is evaluated separately and is subject to the same outcome-independence requirement. A resolution episode is constituted only if at least one role-complete, jointly compatible assignment

of locally admissible witnesses exists. All such assignments are retained, including assignments that induce different terminal behavior.

After constitution, the resolver semantics determines which differences between witness values are relevant. For every resolution-relevant role, we define a contextual equivalence from the complete family of guards and terminal predicates in the declared resolver. Two values are equivalent when substituting one for the other preserves every such predicate in every context of the remaining roles. These role-wise classes are lifted to vectors of local classes and, by a local-to-global lemma, to complete global resolution signatures. A semantic quotient then retains one resolver branch for each distinct global signature rather than one branch for each raw constitutive assignment.

The substitution theorem is formulated at the witness layer. Let a reference realization and a replacement realization instantiate the same resolution contract. A role-preserving map from active replacement witnesses to reference witnesses is required to satisfy three conditions: contextual equivalence of mapped values, preservation of constitutive assignments, and completeness of the local-class vectors realized by the reference system. Together, these conditions imply equality of the realized local-class-vector sets and of the global-signature sets. The corresponding semantic quotient resolvers are therefore canonically isomorphic, and the exact reachable terminal profiles coincide. The conditions are intentionally sufficient rather than necessary: a replacement can preserve the final profile even when this certificate does not apply.

The construction does not propose a new general theory of component substitutability, contextual equivalence, abstraction, or contract realizability. Each of those subjects has an established literature. The contribution lies in the connection between layers: typed heterogeneous witnesses and resolver-independent episode constitution are linked, by a checkable pre-resolver certificate, to exact preservation of a finite resolver. The certificate is evaluated at the interface boundary; global resolver equivalence is a conclusion rather than an assumption.

The paper makes four contributions. First, it defines episode constitution over typed post-transduction witnesses while separating individual admissibility, joint structural compatibility, and resolver semantics. Second, it defines a role-wise, resolver-indexed contextual equivalence from the declared guard and terminal semantics and proves a local-to-global lifting result. Third, it gives sufficient pre-resolver conditions under which heterogeneous-interface substitution preserves the semantic quotient resolver and the exact reachable terminal profile. Fourth, it provides eight

finite adversarial scenarios and a canonical scaling family that test the independence, scope, and computational consequences of the construction.

2 Related Work and Scope

2.1 Interface substitutability and compositional refinement

Interface-aware substitution is an established problem. Interface automata provide a formal basis for compatibility and refinement of component interfaces (de Alfaro and Henzinger 2001). Synchronous relational interfaces distinguish well-formedness or pluggability from substitutability and characterize refinement compositionally for finite-state interfaces (Tripakis et al. 2011). Modal interface theories with data constraints combine interface behavior with explicit data states, pre/postconditions, and guarded input/output transition systems while preserving refinement and compatibility compositionally (Bauer et al. 2011). Later algebraic formulations develop refinement relations expressly designed to preserve component substitutability and include operations such as parallel composition, hiding, conjunction, disjunction, and quotient (Chilton et al. 2014a). Assume-guarantee theories similarly support compositional reasoning for input/output component models and contracts (Chilton et al. 2014b). Work on component-interaction automata gives explicit preconditions under which replacing a component preserves equivalence of the updated system (Černá et al. 2007).

The present result therefore does not claim novelty for conservative component replacement as such. Its substitution map acts at an earlier information boundary: it relates typed witnesses produced by heterogeneous interface realizations before the finite resolver is instantiated. The preservation target is fixed explicitly as the exact set of designated terminal classes reachable in that resolver. The theorem supplies sufficient pre-resolver conditions on witnesses and realized local-class vectors that lift from the interface boundary to preservation of the global resolver.

2.2 Realizability, semantic partitions, and strong preservation

Contract realizability asks whether a contract admits an implementation satisfying its assumptions and guarantees, and formal realizability checking has been developed for theory-rich contract languages (Gacek et al. 2015). Episode constitution addresses a narrower object. It neither

synthesizes nor certifies a complete implementation. Instead, it asks whether the currently available typed witnesses contain at least one role-complete assignment that satisfies the declared local and joint structural checks under a fixed finite resolution contract. Failure of this test occurs before the resolver represented by that contract is instantiated. Relatedly, value interchangeability in constraint satisfaction formalizes when values may be substituted within solutions without losing the relevant solution structure (Freuder 1991). This is close to the structural role of constitutive-assignment preservation, but it does not supply the separate resolver-indexed semantic obligation used below.

Predicate- and observation-induced equivalence classes are likewise well established. Huang, Krafczyk, and Peleska construct equivalence classes over valuations of symbolic finite-state machines from guards, output expressions, and atomic propositions for exhaustive property-oriented model-based testing (Huang et al. 2024). Structured transition systems with parametric observations similarly use observation-dependent bisimulation, congruence conditions, and minimal realizations whose elements characterize congruence classes (Ferrari et al. 1997). The relation used here has a different placement and purpose. It is defined separately for each resolution-relevant input role, before resolver instantiation, and is quantified over every context of the remaining roles because the certified operation is substitution of individual typed witnesses. The resulting role-wise classes are then lifted to complete resolver signatures.

Exact preservation under abstraction also has a substantial theoretical foundation. Strong-preservation results characterize abstractions in which concrete and abstract models satisfy the same formulas of a chosen specification language and relate preservation to simulation, bisimulation, and completeness in abstract interpretation (Ranzato and Tapparo 2007). The semantic quotient below is therefore not presented as a new general abstraction theory. It removes duplicate resolver branches with identical declared guard and terminal semantics. The question addressed here is whether a change made earlier, at the heterogeneous-witness layer, can be certified to leave that quotient unchanged.

Taken together, these lines of work establish the ingredients against which the novelty claim must be delimited: substitutability and refinement of interfaces, well-formedness and realizability, value interchangeability, observational congruence and quotienting, and exact semantic preservation are not claimed as new. The narrower contribution examined here is their placement across a particular boundary: typed post-transduction witnesses are first admitted and jointly constituted

without access to resolver outcomes; resolver-indexed equivalence is then imposed separately; and pre-resolver conditions over witnesses and realized local-class vectors are shown sufficient for exact preservation of the resulting finite resolver. We have not located this combined construction in the literature examined; this statement delimits the search performed and is not a claim of logically exhaustive non-existence.

2.3 Prior dissemination and conceptual lineage

Earlier Spanish-language preprints by the author documented precursor ideas on typed transduction, structured interfaces, observational support, intermodal admissibility, and a broader domain–interface orientation chain (Lloret Egea 2026f,b,e,c,d). They are part of the conceptual and documentary lineage of the present work, but the definitions and proofs below are self-contained. In particular, the 22 June 2026 matrix (Lloret Egea 2026d) does not contain the episode-constitution predicate, the resolver-indexed contextual equivalence, or the substitution theorem proved here.

A separate preprint posted on 8 August 2026 develops exact reachable closure profiles, resolution-covering morphisms, and revision complexity for finite resolution systems (Lloret Egea 2026a). The present paper uses only the elementary reachable-terminal set needed for its preservation result and proves the required invariance self-containedly. A Spanish-language preprint of the present scientific work is also available (Lloret Egea 2026g). It is disclosed as prior dissemination of the same work, not as an independent research contribution on which the present article relies.

3 Mathematical Framework

3.1 Finite resolution contracts

Let

$$\Theta = (D, q, \mathbb{G}) \tag{1}$$

be a *finite resolution contract* for an application domain D and a declared query q . Its guarded resolution scheme is

$$\mathbb{G} = \left(X, \mathcal{A}, E, x_\star, \{g_e\}_{e \in E}, \{t_{v,x}\}_{v \in \{0,1\}, x \in X} \right). \tag{2}$$

Here X is a finite state set, $\mathcal{A}$ is a finite action alphabet, $E \subseteq X \times \mathcal{A} \times X$ is a structural transition relation, and $x_\star$ is the initial state. Each g_e is a Boolean guard. For $v \in \{0, 1\}$, the predicate $t_{v,x}$ determines whether state x belongs to terminal class T_v after valuation.

The scheme is assumed to be in *guarded normal form*: X , $\mathcal{A}$, and E do not depend on the valuation, and every valuation-dependent choice affecting transition availability or membership in either terminal class is represented by a guard or terminal predicate. Parameter-dependent targets or actions must therefore be expanded into guarded structural alternatives.

Let X° be the set of states reachable from $x_\star$ in the underlying directed graph when every guard is replaced by the constantly true predicate. Let E° contain the structural edges whose source lies in X° . For any guard or terminal predicate f , let $\text{supp}(f)$ denote its declared parameter support. Define the conservative structural support

$$P_\Theta^{\text{str}} = \bigcup_{e \in E^\circ} \text{supp}(g_e) \cup \bigcup_{\substack{x \in X^\circ \\ v \in \{0,1\}}} \text{supp}(t_{v,x}). \quad (3)$$

No minimality claim is made for (3). Stronger static analysis may reduce this support; all results below are relative to the declared contract Θ .

For each $p \in P_\Theta^{\text{str}}$, let $\mathcal{V}_p$ be a finite value domain. Every guard and terminal predicate is total on its declared finite product:

$$g_e : \prod_{p \in \text{supp}(g_e)} \mathcal{V}_p \rightarrow \{0, 1\}, \quad (4)$$

$$t_{v,x} : \prod_{p \in \text{supp}(t_{v,x})} \mathcal{V}_p \rightarrow \{0, 1\}. \quad (5)$$

Totality ensures that every hybrid valuation used in the contextual-equivalence construction is defined.

3.2 Typed witnesses and episode constitution

Let W be a finite family of *post-transduction witnesses*. A witness is

$$w = (\text{role}(w), \text{val}(w), \mu(w)), \quad (6)$$

where $\text{role}(w) \in P_{\Theta}^{\text{str}}$, $\text{val}(w) \in \mathcal{V}_{\text{role}(w)}$, and $\mu(w)$ contains the structural metadata required by the contract, such as type, reference, unit, domain, provenance, and declared integrity information. The witness is the output of the declared interface or transduction pipeline at the boundary relevant to the contract. It may arise from one source or from a declared transformation involving several sources. The theory does not identify a witness with a raw sensor reading or with any particular physical technology. For each role p , write

$$W(p) = \{w \in W : \text{role}(w) = p\}. \quad (7)$$

For each role p , a local verifier

$$V_p : W(p) \rightarrow \{0, 1\} \quad (8)$$

checks only the individual admissibility of witnesses assigned to p . It may inspect type, value-domain membership, reference, provenance, and the integrity of the declared acquisition or transduction pipeline. It is independent of resolver semantics: it may not inspect other witness values, the guard or terminal predicates introduced below, terminal membership, a preferred terminal class, or the final reachable profile.

Joint structural compatibility is represented by

$$J(b) = \tilde{J}(\mu \circ b). \quad (9)$$

The metadata projection μ excludes resolver-derived information. In particular, it contains neither local semantic class labels, guard values, terminal values, an expected terminal profile, nor fields computed from those quantities. Equation (9) therefore cannot act as a hidden selector of resolver outcomes.

A *constitutive assignment* is a map

$$b : P_{\Theta}^{\text{str}} \rightarrow W \quad (10)$$

that selects one witness for every required role and satisfies, for every p ,

$$\text{role}(b(p)) = p, \quad V_p(b(p)) = 1, \quad (11)$$

together with $J(b) = 1$. Let $\mathcal{B}_\Theta(W)$ denote the set of all constitutive assignments. Define

$$\boxed{\text{Constit}_\Theta(W) = 1 \iff \mathcal{B}_\Theta(W) \neq \emptyset.} \quad (12)$$

We say that the resolution episode is *constituted under* Θ when (12) holds.

This statement is contract-relative. If $\text{Constit}_\Theta(W) = 0$, the available witness family does not instantiate the resolver specified by Θ . It does not follow that no alternative contract could define a different constituted episode on the same underlying information.

A constitution certificate ζ consists of a candidate constitutive assignment together with the information needed to evaluate the corresponding local-verification and joint-compatibility checks. If every V_p and $\tilde{J}$ is decidable, then

$$\text{Constit}_\Theta(W) = 1 \iff \exists \zeta : \text{VerifyConstit}_\Theta(\zeta) = 1. \quad (13)$$

Failure of this verifier is a pre-resolution failure. It is not assigned a terminal class and is not treated as a property of an already instantiated resolver.

Each $b \in \mathcal{B}_\Theta(W)$ induces a complete valuation

$$\nu_b(p) = \text{val}(b(p)). \quad (14)$$

Different admissible constitutive assignments may disagree. Such disagreement is retained rather than resolved at the constitution layer.

3.3 Resolver-indexed contextual equivalence

Let

$$\mathcal{F}_\Theta = \{g_e : e \in E^\circ\} \cup \{t_{v,x} : v \in \{0,1\}, x \in X^\circ\}. \quad (15)$$

For $p \in P_{\Theta}^{\text{str}}$ and $a \in \mathcal{V}_p$, define the role-wise contextual signature

$$\kappa_p^{\Theta}(a) = (f(\eta[p \mapsto a]))_{\substack{f \in \mathcal{F}_{\Theta}, p \in \text{supp}(f) \\ \eta \in \prod_{q \in \text{supp}(f) \setminus \{p\}} \mathcal{V}_q}}. \quad (16)$$

Thus $\kappa_p^{\Theta}(a)$ records how every resolver predicate that depends on p evaluates when p is set to a , across every possible context of that predicate's remaining arguments.

Define

$$a \equiv_p^{\Theta} b \iff \kappa_p^{\Theta}(a) = \kappa_p^{\Theta}(b). \quad (17)$$

The relation $\equiv_p^{\Theta}$ is an equivalence relation determined by the declared resolver semantics. Let

$$r_p^{\Theta} : \mathcal{V}_p \rightarrow \mathcal{V}_p / \equiv_p^{\Theta} \quad (18)$$

be the quotient map.

For a complete valuation ν , define the global resolution signature

$$\rho_{\Theta}(\nu) = ((g_e(\nu))_{e \in E^{\circ}}, (t_{0,x}(\nu))_{x \in X^{\circ}}, (t_{1,x}(\nu))_{x \in X^{\circ}}). \quad (19)$$

For a constitutive assignment b , define its local class vector

$$\lambda_{\Theta}(b) = \left(r_p^{\Theta}(\nu_b(p)) \right)_{p \in P_{\Theta}^{\text{str}}}, \quad (20)$$

and the realized sets

$$\Lambda_{\Theta}(W) = \{\lambda_{\Theta}(b) : b \in \mathcal{B}_{\Theta}(W)\}, \quad (21)$$

$$\mathcal{S}_{\Theta}(W) = \{\rho_{\Theta}(\nu_b) : b \in \mathcal{B}_{\Theta}(W)\}. \quad (22)$$

Lemma 1 (Local-to-global preservation of resolver semantics). *If two complete valuations ν and ν' satisfy*

$$r_p^{\Theta}(\nu(p)) = r_p^{\Theta}(\nu'(p)) \quad \forall p \in P_{\Theta}^{\text{str}}, \quad (23)$$

then

$$\rho_{\Theta}(\nu) = \rho_{\Theta}(\nu'). \quad (24)$$

Proof. Enumerate $P_{\Theta}^{\text{str}} = \{p_1, \dots, p_k\}$ and construct hybrid valuations $\nu^{(0)} = \nu, \dots, \nu^{(k)} = \nu'$ by replacing one coordinate at a time. At step i , condition (17) states that replacing $\nu(p_i)$ by $\nu'(p_i)$ preserves every $f \in \mathcal{F}_{\Theta}$ that depends on p_i , in every context of the remaining arguments of f . Predicates that do not depend on p_i are unchanged trivially, and totality guarantees that each hybrid context is defined. Hence $f(\nu^{(i-1)}) = f(\nu^{(i)})$ for every $f \in \mathcal{F}_{\Theta}$. Chaining the k substitutions gives equality of every component of (19). $\square$

The converse is not required and need not hold: distinct local class vectors may induce the same global signature.

3.4 Finite resolver and semantic quotient

A constitutive assignment b instantiates

$$\mathcal{R}_{\Theta}[b] = (X, \mathcal{A}, \rightarrow_b, x_{\star}, T_0^b, T_1^b), \quad (25)$$

where $(x, a, y) = e \in E$ gives

$$x \xrightarrow{a}_b y \iff g_e(\nu_b) = 1, \quad (26)$$

and

$$T_v^b = \{x \in X : t_{v,x}(\nu_b) = 1\}. \quad (27)$$

All constitutive assignments are retained in a single finite resolver by introducing a common root $\hat{x}_{\star}$ and one independent branch per assignment. Define

$$\hat{X}_W = \{\hat{x}_{\star}\} \cup (X \times \mathcal{B}_{\Theta}(W)), \quad (28)$$

$$\hat{\mathcal{A}}_W = \mathcal{A} \sqcup \{\iota_b : b \in \mathcal{B}_{\Theta}(W)\}. \quad (29)$$

For every $b \in \mathcal{B}_{\Theta}(W)$,

$$\hat{x}_{\star} \xrightarrow{\iota_b} (x_{\star}, b), \quad (30)$$

and

$$(x, b) \xrightarrow{a} (y, b) \iff x \xrightarrow{a}_b y. \quad (31)$$

The terminal sets are

$$\hat{T}_v = \{(x, b) : x \in T_v^b\}. \quad (32)$$

Denote this system by $\mathcal{O}_\Theta[W]$.

For any finite resolver $\mathcal{R}$, define its exact reachable terminal profile at state x by

$$C_{\mathcal{R}}(x) = \{v \in \{0, 1\} : \text{Reach}_{\mathcal{R}}(x) \cap T_v \neq \emptyset\}. \quad (33)$$

This is the exact set of designated terminal classes reachable from x ; it may be empty, singleton, or $\{0, 1\}$. In the author's separate finite non-closure framework (Lloret Egea 2026a), the same set is termed the *exact reachable closure profile*. No result below depends on any further terminology or semantics from that framework.

Several constitutive assignments may induce the same global signature ρ_Θ . Define the semantic quotient resolver by

$$\hat{X}_W^\star = \{\hat{x}_\star\} \cup (X \times \mathcal{S}_\Theta(W)), \quad (34)$$

$$\hat{\mathcal{A}}_W^\star = \mathcal{A} \sqcup \{\iota_s : s \in \mathcal{S}_\Theta(W)\}. \quad (35)$$

For a signature $s = \rho_\Theta(\nu)$, write s_e for its component associated with guard g_e and $s_{v,x}$ for its component associated with terminal predicate $t_{v,x}$. For every $s \in \mathcal{S}_\Theta(W)$,

$$\hat{x}_\star \xrightarrow{\iota_s} (x_\star, s). \quad (36)$$

For $e = (x, a, y) \in E^\circ$,

$$(x, s) \xrightarrow{a} (y, s) \iff s_e = 1. \quad (37)$$

Finally,

$$\hat{T}_v^\star = \{(x, s) : x \in X^\circ, s_{v,x} = 1\}. \quad (38)$$

Denote this quotient by $\mathcal{O}_\Theta^\star[W]$.

Lemma 2 (Terminal-profile invariance under the semantic quotient).

$$C_{\mathcal{O}_\Theta[W]}(\hat{x}_\star) = C_{\mathcal{O}_\Theta^*[W]}(\hat{x}_\star). \quad (39)$$

Proof. Constitutive assignments with the same global signature induce the same enabled structural transitions and the same terminal memberships, and therefore the same branch profile. The quotient retains one branch for each distinct signature. Removing duplicate branches cannot add or remove a reachable terminal class from the common root. $\square$

3.5 Profile-preserving substitution of heterogeneous interfaces

Define the active witness set

$$W_{\text{act}} = \{w \in W : \exists b \in \mathcal{B}_\Theta(W), \exists p, b(p) = w\}, \quad (40)$$

and let $W_{\text{act}}(p) = W_{\text{act}} \cap W(p)$. Only witnesses that occur in at least one constitutive assignment are active.

Let W be the reference realization and W' a replacement realization of the same contract Θ , with both episodes constituted. A typed substitution consists of maps

$$m_p : W'_{\text{act}}(p) \rightarrow W(p) \quad (41)$$

for every $p \in P_\Theta^{\text{str}}$. The direction of (41) is deliberate: every active witness produced by the replacement is paired with a role-matched reference witness against which preservation is checked. For $b' \in \mathcal{B}_\Theta(W')$, define the mapped assignment by

$$(m \circ b')(p) = m_p(b'(p)). \quad (42)$$

Three sufficient conditions are imposed.

C2—Local contextual equivalence.

$$\text{val}(w') \equiv_p^\Theta \text{val}(m_p(w')) \quad \forall w' \in W'_{\text{act}}(p). \quad (43)$$

C3—Constitutive-assignment preservation.

$$b' \in \mathcal{B}_\Theta(W') \implies m \circ b' \in \mathcal{B}_\Theta(W). \quad (44)$$

C4_λ—Completeness of realized local-class vectors.

$$\forall \lambda \in \Lambda_\Theta(W), \exists b' \in \mathcal{B}_\Theta(W') : \lambda_\Theta(b') = \lambda. \quad (45)$$

These conditions are conservative and sufficient. They are not claimed to characterize all substitutions that preserve (33).

Lemma 3 (Equality of realized local-class-vector sets). *Under C2 and C3,*

$$\Lambda_\Theta(W') \subseteq \Lambda_\Theta(W). \quad (46)$$

If C4_λ also holds, then

$$\Lambda_\Theta(W') = \Lambda_\Theta(W). \quad (47)$$

Proof. Take any $b' \in \mathcal{B}_\Theta(W')$. By C3, $m \circ b' \in \mathcal{B}_\Theta(W)$. By C2, each coordinate of b' belongs to the same contextual class as the corresponding coordinate of $m \circ b'$. Hence $\lambda_\Theta(b') = \lambda_\Theta(m \circ b') \in \Lambda_\Theta(W)$, which proves (46). Condition C4_λ supplies the reverse inclusion. $\square$

Lemma 4 (Local-class-vector equality implies global-signature equality). *If (47) holds, then*

$$\mathcal{S}_\Theta(W') = \mathcal{S}_\Theta(W). \quad (48)$$

Proof. Take a signature induced by some $b' \in \mathcal{B}_\Theta(W')$. Equality of the realized local-class-vector sets provides $b \in \mathcal{B}_\Theta(W)$ with $\lambda_\Theta(b') = \lambda_\Theta(b)$. Lemma 1 then gives $\rho_\Theta(\nu_{b'}) = \rho_\Theta(\nu_b)$. This proves one inclusion between the signature sets; the reverse inclusion follows symmetrically. $\square$

Theorem 1 (Pre-resolver conditions for exact terminal-profile preservation). *Let W and W' be constituted heterogeneous-interface realizations of the same finite resolution contract Θ . If a typed*

substitution (41) satisfies C2, C3, and $C4_\lambda$, then

$$\mathcal{S}_\Theta(W') = \mathcal{S}_\Theta(W), \quad (49)$$

the semantic quotient resolvers are canonically isomorphic,

$$\mathcal{O}_\Theta^*[W'] \cong \mathcal{O}_\Theta^*[W], \quad (50)$$

and

$$\boxed{C_{\mathcal{O}_\Theta[W']}(\hat{x}_\star) = C_{\mathcal{O}_\Theta[W]}(\hat{x}_\star)}. \quad (51)$$

Proof. The preceding lemmas give equality of the realized global-signature sets. Both quotient resolvers are constructed from the same structural state set and contain one branch for each realized signature. Identifying the two common roots and mapping every state (x, s) to its counterpart with the same pair (x, s) defines a canonical bijection. This bijection preserves and reflects every initial branch selection, enabled internal transition, and terminal membership, and is therefore an isomorphism of the two finite resolvers. Consequently, a terminal class is reachable from the root of one quotient resolver if and only if the corresponding class is reachable from the root of the other. Lemma 2 transfers this equality to the unquotiented resolvers, proving (51). $\square$

Condition $C4_\lambda$ quantifies only over vectors of local contextual classes. It does not quantify over global signatures, complete resolvers, terminal outcomes, or the final profile. The bridge from local-class vectors to the global-signature set is supplied by Lemmas 1 and 4.

3.6 Limits of the certificate and an indistinguishability obstruction

Failure of C2– $C4_\lambda$ does not establish a change in the terminal profile. Two realizations may have different global signature sets while reaching exactly the same designated terminal classes:

$$\mathcal{S}_\Theta(W') \neq \mathcal{S}_\Theta(W) \quad \text{and yet} \quad C_{\mathcal{O}_\Theta[W']}(\hat{x}_\star) = C_{\mathcal{O}_\Theta[W]}(\hat{x}_\star). \quad (52)$$

The theorem therefore provides a verifiable sufficient certificate, not a necessary characterization.

Let

$$\Omega_{D,\Theta}^{\text{con}} = \{\omega : \text{Constit}_\Theta(W(\omega)) = 1\}. \quad (53)$$

For each $\omega \in \Omega_{D,\Theta}^{\text{con}}$, write

$$C(\omega) = C_{\mathcal{O}_\Theta[W(\omega)]}(\hat{x}_\star). \quad (54)$$

Proposition 1 (Indistinguishability obstruction). *Let $\pi_S : \Omega_{D,\Theta}^{\text{con}} \rightarrow Y_S$ represent all information made available by an interface family S . If there exist $\omega_1, \omega_2 \in \Omega_{D,\Theta}^{\text{con}}$ such that*

$$\pi_S(\omega_1) = \pi_S(\omega_2), \quad C(\omega_1) \neq C(\omega_2), \quad (55)$$

then no deterministic function $F : Y_S \rightarrow \mathcal{P}(\{0, 1\})$ can recover the exact terminal profile for both states.

Proof. The function F receives the same input for ω_1 and ω_2 and must therefore return the same output, whereas the correct terminal profiles differ. $\square$

3.7 Computational scope

All value domains and resolvers considered here are finite. Exhaustive construction of the role-wise contextual equivalences has worst-case evaluation cost bounded by

$$O\left(\sum_{p \in P_\Theta^{\text{str}}} |\mathcal{V}_p|^2 \sum_{\substack{f \in \mathcal{F}_\Theta \\ p \in \text{supp}(f)}} \left[\prod_{q \in \text{supp}(f) \setminus \{p\}} |\mathcal{V}_q| \right] c_f\right), \quad (56)$$

where c_f is the evaluation cost of f . The cost may therefore grow exponentially with predicate support arity. The theorem does not assume an inexpensive equivalence checker. Restricted predicate languages may instead admit symbolic SAT- or SMT-based procedures.

4 Reproducible Finite Evaluation

The evaluation has two purposes. The first is adversarial: the scenarios separate the functions of the theorem’s hypotheses and test semantic distinctions that could otherwise be conflated. The second is structural: the scaling family shows how quotienting by declared resolver semantics can

remove large multiplicities of raw constitutive assignments without changing the exact terminal profile. Every example is finite and can be checked exhaustively. The supplementary deterministic verifier implements the finite contracts, local witness verification, joint structural compatibility, constitutive-assignment enumeration, contextual equivalence, local and global signatures, raw and quotient resolvers, exact reachability profiles, and conditions C2–C4_λ.

4.1 Common resolver

Seven scenarios use the common state set

$$X = \{x_*, a, b, c, d, t_0, t_1\}, \quad (57)$$

with $p \in \{0, 1, 2, 3\}$ and $q, r \in \{0, 1\}$. The guarded edges are

$$x_* \rightarrow a \iff r = 1, \quad x_* \rightarrow b \iff q = 1, \quad (58)$$

$$a \rightarrow c \iff q = 1, \quad b \rightarrow c \iff r = 1, \quad (59)$$

$$c \rightarrow t_0 \iff p \in \{0, 2\}, \quad c \rightarrow t_1 \iff p = 1, \quad (60)$$

$$c \rightarrow d \iff p = 3, \quad d \rightarrow t_0 \text{ unconditionally.} \quad (61)$$

The terminal classes are $T_0 = \{t_0\}$ and $T_1 = \{t_1\}$. With $q = r = 1$,

$$p \in \{0, 2, 3\} \Rightarrow C = \{0\}, \quad p = 1 \Rightarrow C = \{1\}. \quad (62)$$

However,

$$0 \equiv_p^\Theta 2, \quad 0 \not\equiv_p^\Theta 1, \quad 0 \not\equiv_p^\Theta 3, \quad (63)$$

because $p = 0$ and $p = 3$ induce different guard values: the former enables $c \rightarrow t_0$, whereas the latter enables $c \rightarrow d$. Both branches eventually reach t_0 , but contextual equivalence records the declared guard and terminal semantics rather than terminal reachability alone. Scenario F2 uses the coupled guard $g(p, q) = [p + q > 5]$ to distinguish contextual from pointwise agreement.

Table 1: Adversarial finite scenario suite.

ID	Target	Construction	Reference	Replacement
F1	Independence of C2	W realizes $p = 0$; W' realizes $p = 0, 1$. The typed map sends both replacement values to the reference witness with $p = 0$. C3 and $C4_\lambda$ hold; C2 fails on $1 \mapsto 0$.	$\{0\}$	$\{0, 1\}$
F2	Contextual versus pointwise equivalence	Guard $g = [p + q > 5]$. Reference $p = 2$, replacement $p = 3$, with admissible $q = 3, 4$. The two p values agree when $q = 4$ but disagree when $q = 3$.	$\{0, 1\}$	$\{1\}$
F3	Independence of C3	Reference W has valid $p = 0$ and a locally admissible $p = 1$ witness with the same resolver value as its replacement counterpart but incompatible episode metadata. Replacement W' has valid $p = 0, 1$. C2 and $C4_\lambda$ hold, but the image of the $p = 1$ assignment is not constitutive.	$\{0\}$	$\{0, 1\}$
F4	Independence of $C4_\lambda$	W realizes local classes for $p = 0$ and $p = 1$; W' retains only the $p = 0$ class. C2 and C3 hold for every surviving active witness.	$\{0, 1\}$	$\{0\}$
F5	Why the map is restricted to W'_{act}	Add a locally admissible witness that cannot occur in any constitutive assignment because its metadata fail J. Requiring a substitution image for it would cause the certificate to fail although no resolver branch changes.	$\{0\}$	$\{0\}$
F6	Why the primitive is a post-transduction witness	Two independently admissible transduction pipelines produce distinct witnesses for the same role, with values 0 and 1. Prematurely merging them removes a legitimate constitutive branch.	$\{0, 1\}$	$\{0, 1\}$
F7	Replacement without raw-value equality	The reference interface yields $p = 0$ and the replacement yields $p = 2$. The raw values differ, but $0 \equiv_p^\Theta 2$; C3 and $C4_\lambda$ hold.	$\{0\}$	$\{0\}$
F8	Sufficiency is not necessity	The reference realization uses $p = 0$ and a direct edge $c \rightarrow t_0$; the replacement uses $p = 3$ and the path $c \rightarrow d \rightarrow t_0$. Local classes and signatures differ, but the reachable terminal profiles coincide.	$\{0\}$	$\{0\}$

4.2 Interpretation of the adversarial suite

The scenarios establish different facts rather than repeating a single ablation. F1 shows that C2 is not redundant with role typing, assignment preservation, and local-class-vector completeness. F2 rules out the weaker condition in which C2 is tested only in the currently observed context. Because the replacement must remain semantically interchangeable wherever the declared resolver may place that role value, pointwise agreement at one context is insufficient.

F3 isolates C3. Equality of resolver-relevant values does not repair a structurally inadmissible assignment: provenance, reference, or another declared metadata constraint can prevent the mapped witnesses from forming a legitimate episode. F4 isolates $C4_\lambda$. Conditions C2 and C3 prevent the replacement from introducing local semantic classes that are absent from the reference realization, but they do not prevent the replacement from losing a reference class. Completeness supplies precisely that reverse coverage.

F5 shows why inactive witnesses are outside the substitution obligation. A witness that can participate in no constitutive assignment contributes no branch to the resolver. F6 shows why the interface primitive is placed after declared transduction rather than at the raw-source level: several heterogeneous sources can feed one witness, and distinct admissible transductions can yield conflicting witnesses that must remain separate until the resolver evaluates them. F7 confirms that the theorem does not require physical identity or raw-value equality. F8 marks the opposite boundary: failure of the sufficient conditions does not imply that the reachable terminal profile has changed.

4.3 Canonical scaling family

To isolate representational multiplicity from resolver semantics, consider k roles with value domain $\{0, 1, 2, 3\}$ and predicates that distinguish only parity in each role. Then

$$\{0, 2\} \quad \text{and} \quad \{1, 3\} \tag{64}$$

are the two contextual classes for each role. If every raw value is represented independently, the number of constitutive assignments is 4^k , whereas the number of local class vectors is 2^k . A guard family exposing each parity bit yields one global signature per local class vector. Table 2 gives the

exact counts.

Table 2: Exact quotient counts in the canonical scaling family.

k	Raw assignments 4^k	Signatures 2^k	Reduction
2	16	4	$4\times$
3	64	8	$8\times$
4	256	16	$16\times$
5	1,024	32	$32\times$
6	4,096	64	$64\times$
7	16,384	128	$128\times$
8	65,536	256	$256\times$

These are exact combinatorial counts, not runtime measurements. They quantify the reduction obtained when multiple constitutive assignments are indistinguishable to every declared guard and terminal predicate. By Lemma 2, this reduction leaves the reachable terminal profile unchanged.

5 Discussion

5.1 Constitution is distinct from resolution

Equation (12) separates three logically different situations. First, an individual interface output can fail its local verifier; that witness cannot support its assigned role. Second, every available witness may be individually admissible while no role-complete set of witnesses satisfies the joint structural constraints; in that case the episode is not constituted under Θ . Third, a constituted resolver may have an empty, singleton, or two-class reachable terminal profile. Only the third case is a property of an instantiated resolver.

This distinction prevents a category error. Failure to instantiate the declared resolution problem is not a negative terminal result. Conversely, successful constitution does not imply a unique terminal class; it establishes only that the declared finite resolver has been instantiated from admissible interface evidence.

5.2 Admissible disagreement is retained

The factorization in (9) and the independence of the local verifiers from resolver semantics are deliberate constraints. They prevent the constitution layer from filtering witnesses because of the resolver outcome they would induce. If two individually admissible witnesses disagree but each

participates in a valid constitutive assignment, both assignments remain in $\mathcal{B}_\Theta(W)$ and therefore both branches remain in $\mathcal{O}_\Theta[W]$. The resolver, not the interface-admission layer, determines the consequence of that disagreement.

This separation is particularly important when interfaces are heterogeneous. The same role can be supported by different acquisition or transformation pipelines that have different physical implementations, data provenance, or numerical encodings. The theorem does not require those realizations to be identical. It requires preservation of the resolver-relevant contextual class together with preservation and coverage of the constitutive structure.

5.3 Relationship to established substitution and preservation theories

The theorem is deliberately narrower than general substitutability results. Interface automata, relational interfaces, interface theories with data, assume–guarantee reasoning, and component-substitutability frameworks already provide broad accounts of compatibility, refinement, and safe component replacement (de Alfaro and Henzinger 2001; Tripakis et al. 2011; Bauer et al. 2011; Chilton et al. 2014a,b; Černá et al. 2007). Likewise, value interchangeability, observational congruence, semantic quotienting, and strong preservation have established theories that are not reproduced here (Freuder 1991; Ferrari et al. 1997; Ranzato and Tapparo 2007). The present construction fixes a specific information boundary: the replacement is expressed over typed witnesses that precede a finite resolver, and the certificate must be checkable without assuming the global resolver equivalence that it is intended to establish.

This placement also explains the role of $C4_\lambda$. The condition does not require equality of global signatures directly. Instead, it requires the replacement to realize every local-class vector present in the reference realization. Combined with C2 and C3, it yields equality of the realized local-class-vector sets; Lemma 1 then lifts that equality to global signatures. The theorem therefore provides a sufficient certificate connecting a pre-resolver witness layer to an exact global invariant.

The result should not be read as a necessary characterization. F8 gives a finite counterexample: two realizations may differ in local classes and in transition structure while still reaching the same terminal class. A failed certificate therefore means only that the present sufficient route has not established preservation. It is not evidence that the replacement changed the outcome.

5.4 Applicability beyond a particular interface technology

Nothing in the mathematics requires a witness to originate from a deterministic physical sensor. A witness may be produced by measurement hardware, a software service, a database, a rule-based transformation, or a learned model, provided that the application supplies the declared local admissibility and metadata checks. The theorem does not validate those domain-specific checks. It only constrains their formal role and shows what follows when they are satisfied.

The same distinction applies to uncertainty. Statistical or probabilistic procedures may be used upstream to generate a candidate witness. The finite constitution and resolver semantics studied here remain conditional on the resulting declared witness values and verification rules. No probability model is required by the theorem, and no claim is made that probabilistic uncertainty can always be reduced to the finite domains assumed here.

6 Limitations

The framework is finite. Continuous dynamics, unbounded value domains, or partially defined guard semantics require a finite abstraction or a different theory. The structural support in (3) is conservative and may require witnesses for structurally reachable branches that a stronger static analysis could prove irrelevant. Consequently, $\text{Constit}_\Theta(W) = 0$ is a statement about the declared contract, not an impossibility theorem over all alternative formulations of the problem.

The domain-specific correctness of the local verifiers, metadata model, and joint compatibility relation is treated extensionally. The framework constrains what these mechanisms may inspect so that they cannot encode the desired resolver outcome covertly, but it does not establish that a particular calibration rule, provenance rule, unit conversion, database record, or learned-model validation criterion is scientifically correct. Those judgments remain external obligations of the application domain.

The contextual-equivalence construction may be expensive. Equation (56) shows exponential growth in predicate support arity in the worst case. Symbolic methods may reduce the cost for suitable predicate languages, but no polynomial-time claim is made here.

Finally, C2, C3, and $C4_\lambda$ are sufficient but not necessary for equality of the reachable terminal profile. The theorem therefore supplies a conservative certification route rather than a complete

classification of all profile-preserving substitutions. The evaluation is formal and computational: it exhaustively checks the finite scenarios and the scaling family, but it does not substitute for domain-specific empirical validation of a concrete interface technology.

7 Conclusion

This paper has developed a finite-systems framework that separates episode constitution from resolver execution under heterogeneous interfaces. A finite resolution contract determines the roles and structural support relevant to its guard and terminal semantics. Typed post-transduction witnesses are checked locally without access to resolver outcomes, and jointly compatible role-complete selections form constitutive assignments. Legitimate disagreement can therefore remain explicit instead of being eliminated by an upstream fusion rule.

Resolver-indexed contextual equivalence identifies, for each role, the finite values that the declared guard and terminal semantics cannot distinguish in any context of the remaining roles. The local-to-global lemma lifts equality of the resulting role-wise classes to equality of complete resolver signatures. On that basis, C2, C3, and $C4_\lambda$ provide a sufficient pre-resolver certificate for heterogeneous-interface substitution: they imply equality of realized local-class-vector sets, equality of global-signature sets, canonical isomorphism of the semantic quotient resolvers, and exact preservation of the reachable terminal profile.

The adversarial suite delineates the scope of the certificate. Local contextual equivalence, constitutive-assignment preservation, and local-class-vector completeness protect different failure modes; inactive witnesses need not be mapped; raw-value equality is unnecessary; and certificate failure does not imply a change in the terminal profile. The scaling family further shows that the semantic quotient can remove large multiplicities of raw constitutive assignments while preserving exact terminal behavior. The result is a finite, checkable connection between heterogeneous interface realizations and a global resolver invariant, not a new general theory of component substitutability or abstraction.

Data availability

No data were generated or analyzed in this study.

Code availability

A journal-independent deterministic Python verifier reproducing the eight finite adversarial scenarios and the canonical scaling family has been submitted to Code Ocean for archival publication under the title *Finite Resolution Systems: Deterministic Substitution Verifier and Adversarial Suite* (provisional DOI: <https://doi.org/10.24433/CO.5078971.v1>). The mathematical proofs in this article are self-contained and do not depend on the verifier.

Funding

No external funding was received for this work.

Competing interests

Competing interests: The author declares none.

Declaration of generative AI use

OpenAI ChatGPT (GPT-5.6 Sol), accessed through the ChatGPT web application during August 2026, was used as a research-support tool for literature discovery, adversarial consistency checking, manuscript structuring, assistance with mathematical formalisation and typesetting support, and English-language editing. Grok 4.5 (xAI), accessed through the Grok web interface on 8 August 2026, was used as a research-support tool for critical review of successive manuscript versions, identification of structural and mathematical improvements, and suggestions on presentation and positioning. Neither tool was modified or trained by the author for this work. All AI outputs were treated as non-authoritative research inputs. No AI-generated figures, datasets, or experimental results are used in the manuscript. The author established, independently checked, reviewed, and approved all definitions, mathematical claims, proofs, interpretations, references, and conclusions and takes full responsibility for the submitted manuscript.

Biographical note

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